FINAL CCNA GRADUATION PROJECT

SecureCorp
Multi-Site Enterprise Network

A two-site network design featuring VLAN segmentation, multi-area OSPF routing, and full Layer 2/3 security hardening — built and configured in Cisco Packet Tracer.

Mohamed Ali
Network & Security Configuration

Ahmed Alaa
Network & Security Configuration

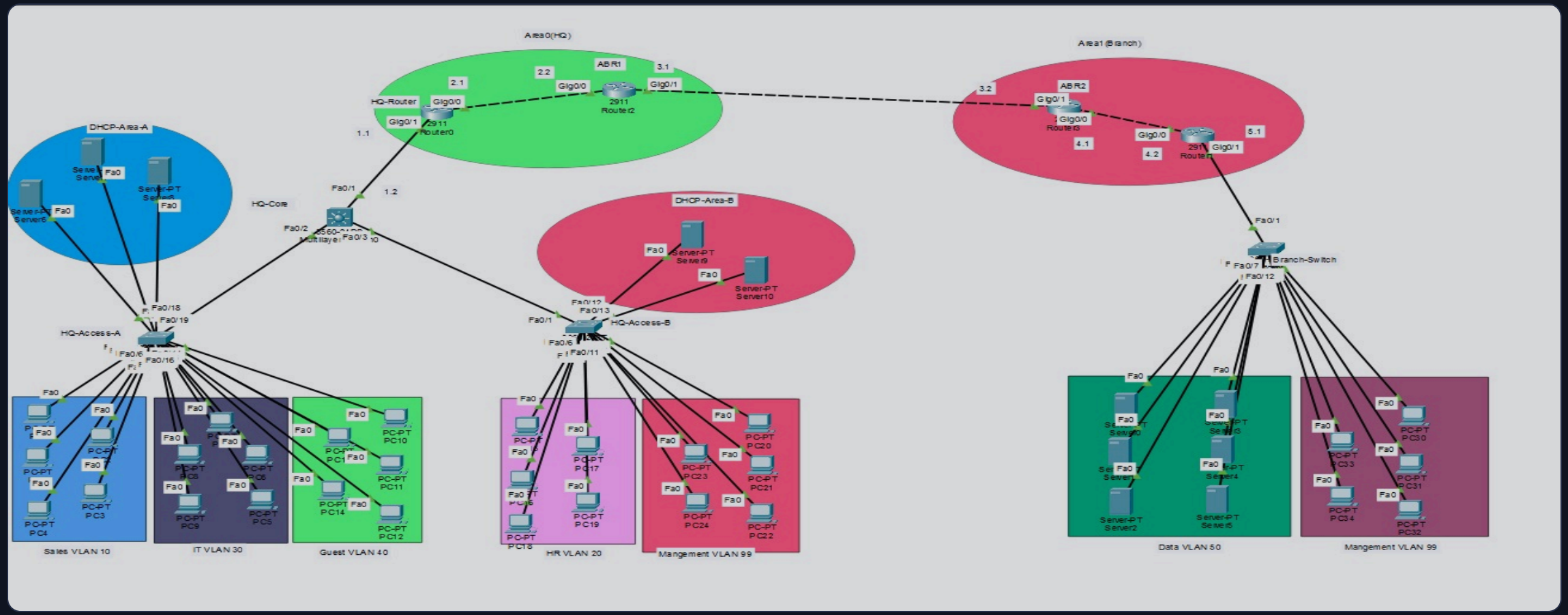
OVERVIEW

Project Summary

- **Two sites:** Headquarters (HQ) and a Branch office, linked over a multi-router backbone.
- **Five VLANs at HQ:** Sales, HR, IT, Guest, and Management, each with its own subnet and DHCP pool.
- **Two VLANs at Branch:** Data and Management, routed via a dedicated Branch router.
- **Multi-area OSPF:** Area 0 as the HQ backbone, Area 1 for the Branch site.
- **Full security hardening:** port security, DHCP snooping, BPDU/Root Guard, ACLs, and SSH-based remote management.

DESIGN

Network Topology



ROUTING

Multi-Area OSPF Design

DEVICE	INTERFACES	OSPF AREA
HQ-Router	Gi0/0 → ABR1 (192.168.2.1), Gi0/1 → HQ-Core (192.168.1.1)	0
ABR1	Gi0/0 ← HQ-Router (192.168.2.2), Gi0/1 → ABR2 (192.168.3.1)	0
ABR2	Gi0/1 ← ABR1 (192.168.3.2), Gi0/0 → Branch-Router (192.168.4.1)	0 / 1 (border)
Branch-Router	Gi0/0 ← ABR2 (192.168.4.2), Gi0/1 → Branch-Switch + VLAN sub-interfaces	1

HQ-Core also participates in OSPF Area 0, advertising all five HQ VLAN subnets from its Layer 3 SVIs.

ADDRESSING

VLAN & IP Plan

VLAN	NAME	SITE	SUBNET
10	Sales	HQ	192.168.10.0/24
20	HR	HQ	192.168.20.0/24
30	IT	HQ	192.168.30.0/24
40	Guest	HQ	192.168.40.0/24
99	Management	HQ	192.168.99.0/24
50	Data	Branch	172.16.50.0/24
99	Management	Branch	172.16.99.0/24
666	Black-hole (unused ports)	Both	not routed

LAYER 2 SECURITY

Switching & Attack Mitigation

- **Port Security** Sticky MAC learning on every access port, violation mode set to `restrict`.
- **BPDU Guard** Enabled on all access ports to block rogue switches.
- **Root Guard** Enabled on HQ-Access-A's uplink (Fa0/24) to protect the STP root.
- **VLAN 666** All unused ports parked in a black-hole VLAN and administratively shut down.
- **DHCP Snooping** Enabled on HQ-Core and Branch-Switch, with uplinks trusted.

SERVICES & ACCESS CONTROL

DHCP, ACLs & Remote Management

DHCP

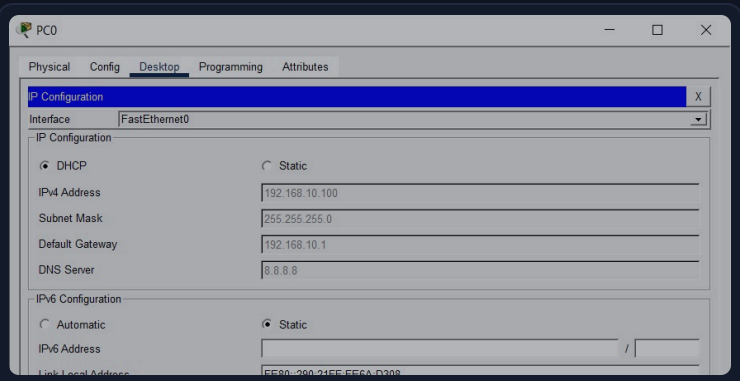
- Five pools on HQ-Core: Sales, HR, IT, Guest, Management
- Two pools on Branch-Router: Data, Management
- Gateway addresses excluded from every pool

Access Control

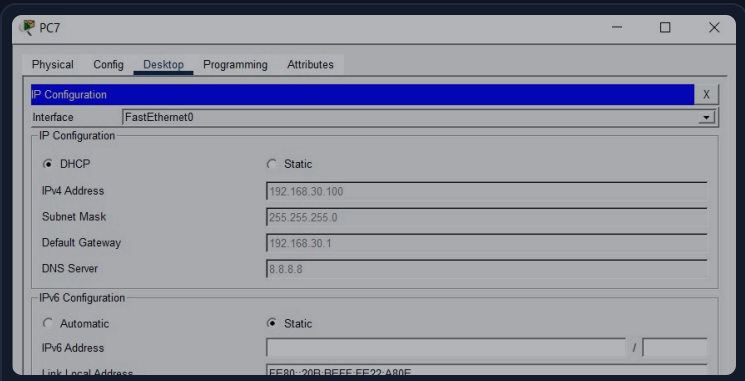
- `Guest-Restrict` ACL blocks Guest VLAN from all internal VLANs
- SSH-only management on HQ-Router with local login
- Enable secrets encrypted; login banners configured on every device

VERIFICATION

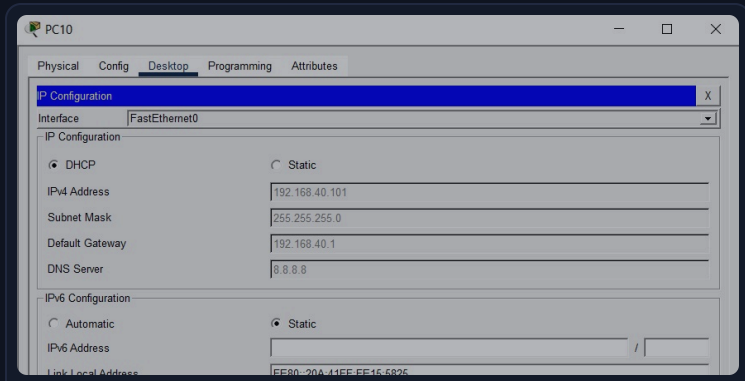
DHCP Working Correctly Per VLAN



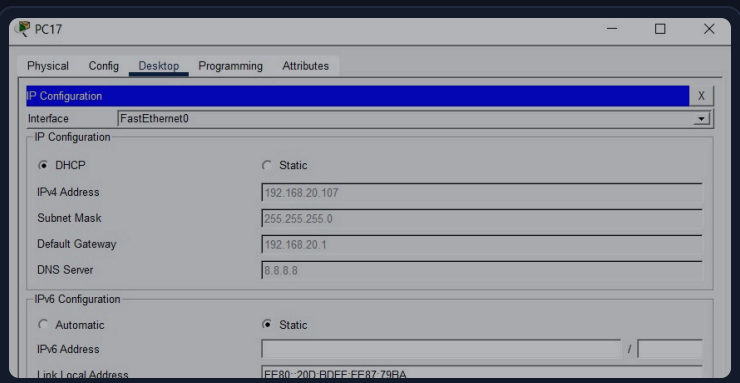
PC0 — Sales VLAN 10 — 192.168.10.100



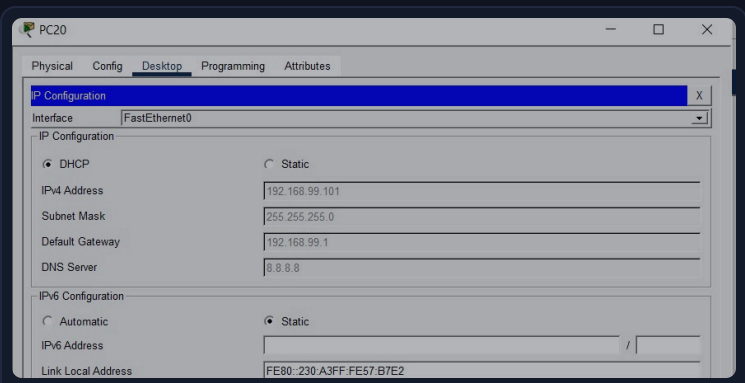
PC7 — IT VLAN 30 — 192.168.30.100



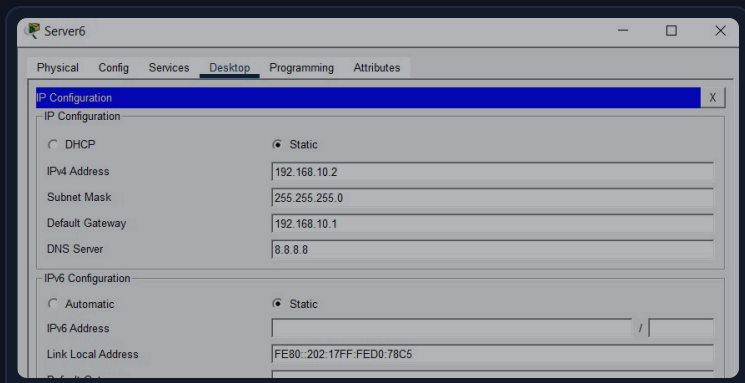
PC10 — Guest VLAN 40 — 192.168.40.101



PC17 — HR VLAN 20 — 192.168.20.107



PC20 — Management VLAN 99 — 192.168.99.101



Server6 — Sales VLAN — Static 192.168.10.2

CONCLUSION

Project Team

- Designed and configured a complete two-site enterprise network with VLAN segmentation, multi-area OSPF, and full security hardening.
- Verified end-to-end functionality: DHCP addressing, inter-VLAN routing, and access restrictions all confirmed working as designed.

Mohamed Ali
Project Team Member

Ahmed Alaa
Project Team Member